simulation. Periodic boundary conditions can be used to ensure constant volume. The density function may then be represented by a Dirac delta function as follows:

$$\rho_{\mathrm{mc}} = Z^{-1}\delta[H - \mathrm{E}], \qquad Z = \int_{\mathcal{D}} \delta[H - \mathrm{E}]\mathrm{d}\omega.$$

The partition function Z, which normalizes the density, is effectively a function of N, V and E; it represents the number of microstates available under given conditions. As this ensemble is associated to constant particle number N, volume V and energy E, it is often referred to as the NVE-ensemble, and when we speak of NVE simulation, we mean simulation that is meant to preserve the microcanonical distribution; this, most often, would be based on approximating Hamiltonian dynamics, e.g. using the Verlet method or another of the methods introduced in Chaps. 2 and 3, and assuming the ergodic property. For a discussion of alternative stochastic microcanonical methods see [126].

6.1.2 *The Canonical Ensemble and the Boltzmann-Gibbs Distribution*

Next, we consider the formulation of an ensemble model that will better capture the realistic environment of a molecular system. Let the system of interest embedded in a much larger system, which acts as a "heat bath." The two systems are assumed to be weakly coupled, so energy is continuously exchanged between them. This energy is assumed to be in the form of heat, with equilibrium naturally associated to the systems having matching temperature. The sole mechanism of interaction between the two systems is in the form of energy exchange, they do not exchange particles (or momentum).

The cornerstone of statistical mechanical theory is the concept of entropy, which may be taken as the number of states available to a discrete system (e.g. a quantum mechanical system), or, in the case of a classical system, the volume of the accessible phase space. In terms of a microcanonical system, the entropy is enshrined in the famous formula that appears on Boltzmann's tombstone:[2]

$$S \equiv \mathrm{k_B}\ \ln Z, \tag{6.1}$$

where $\mathrm{k_B}$ is Boltzmann's constant $\mathrm{k_B} \approx 1.38 \times 10^{-23}$ J/K. The second law of thermodynamics states that the entropy in a closed system naturally tends to increase to a maximum (a critical point).

[2]Despite Boltzmann being the first to formulate the concept, it was Planck who first wrote this equation.